MACH Lab Overview

Multiphysics Aerothermodynamics & Computational Hypersonics Laboratory at Purdue University

Principal Investigator: Prof. Kyle Hanquist | School of Aeronautics and Astronautics | Purdue University

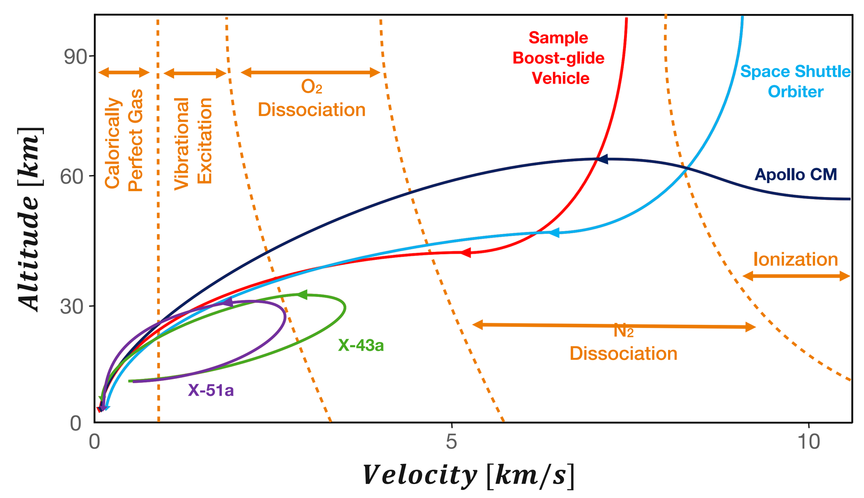
Who We Are

The MACH Lab develops and applies advanced physical models and computational methods to investigate the complex, multiphysics environments encountered in hypersonic flight and atmospheric entry. Using computational modeling and high-performance computing, we seek to improve the physical understanding and predictive capability needed to manage extreme heating and other coupled phenomena, enabling safer and more capable systems for national security and space exploration.

Our Focus

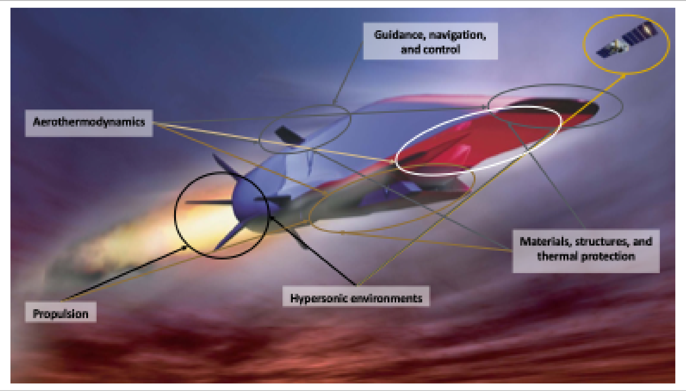
Understanding and predicting the coupled physics of hypersonic flight and atmospheric entry, and using advanced computational modeling to extend beyond the limits of experiments and enable safer, more capable aerospace systems.

Where High-Temperature & Real Gas Effects Matter



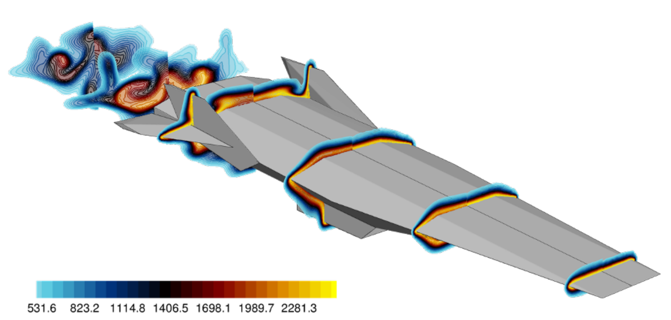
At higher flight energies, chemical kinetics increasingly govern heating, nonequilibrium flow, plasma formation, and material response.

What We Study



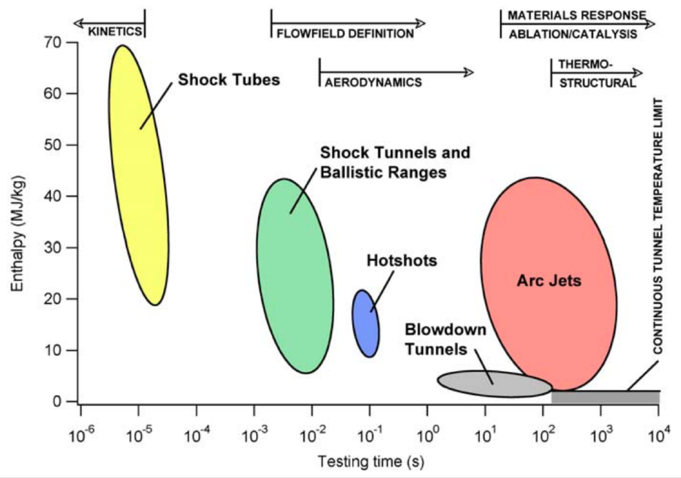
Multi-Physics Nature of Hypersonics

Predicting shocks, aerodynamic loads, boundary layers, and surface heating in high-enthalpy flight environments.



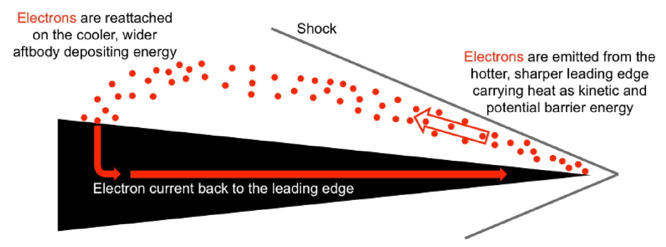
Hypersonic Aerothermodynamics, Design & Optimization

Modeling aerodynamics, heating, and flow–vehicle interactions to support hypersonic vehicle design, optimization, and performance prediction.



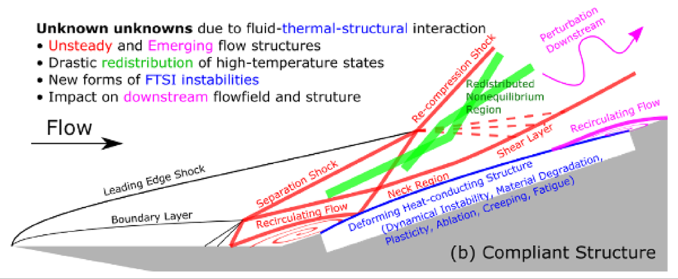
Validation Across Experimental Scales

No single experiment can reproduce the full enthalpy and time scales of hypersonic flight. We validate our models across complementary experimental regimes and use them to explore conditions beyond the reach of any one facility. Image from NASA.



Novel Thermal Management Techniques

Exploring electron transpiration cooling, where thermionically emitted electrons remove energy from hot surfaces and interact with the surrounding hypersonic plasma.

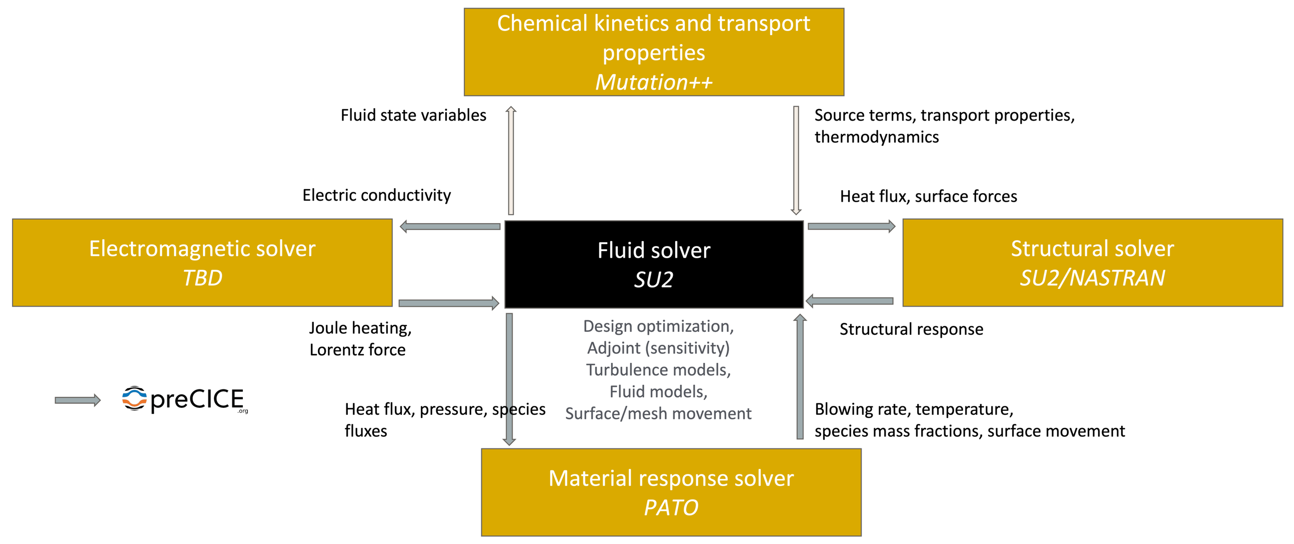


Fluid-Structure Interaction

Investigating the coupling between hypersonic aerodynamic and thermal loading and the structural response of compliant surfaces.

How We Do It

Multi-physics computational workflow

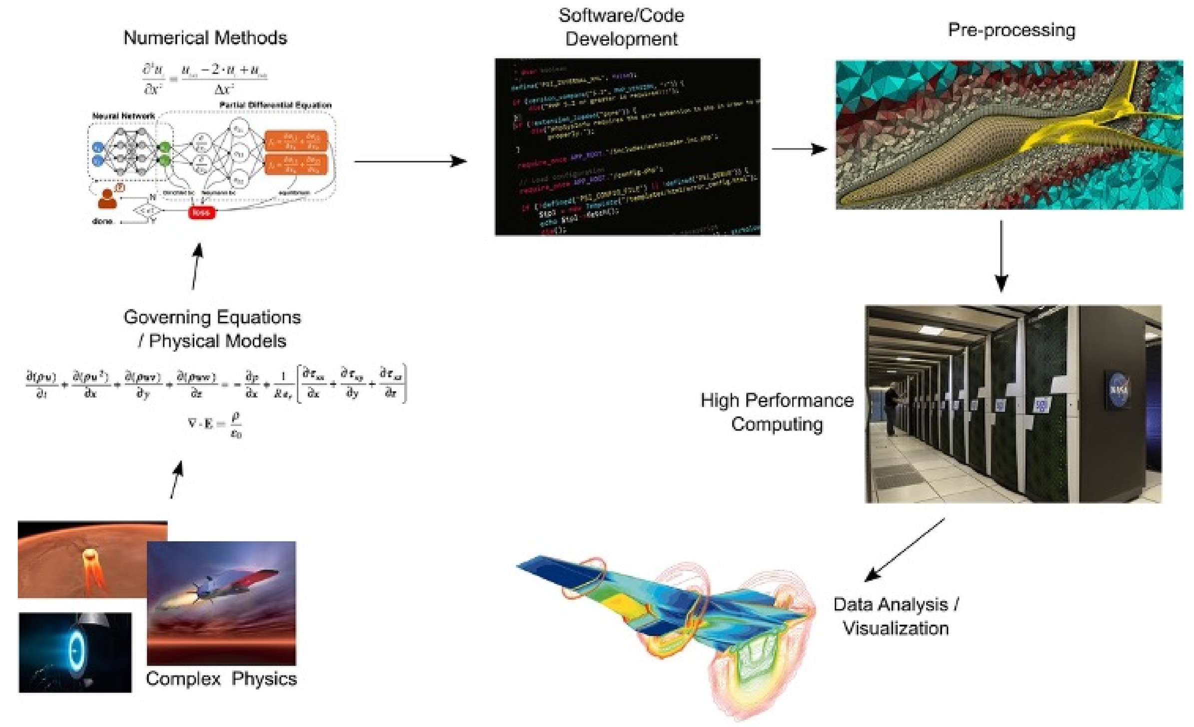


Software & Modeling Tools

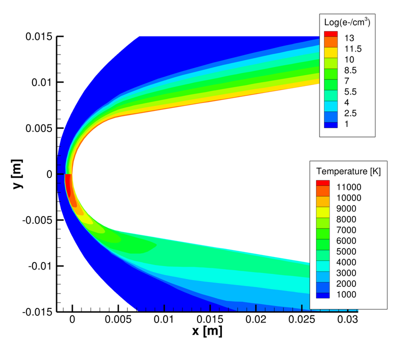
SU2-NEMO | Mutation++ | WarpX | PATO | ChemGen

Pointwise | Link3D | Dakota | preCICE

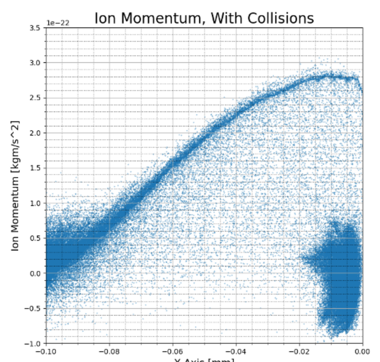
Our Workflow



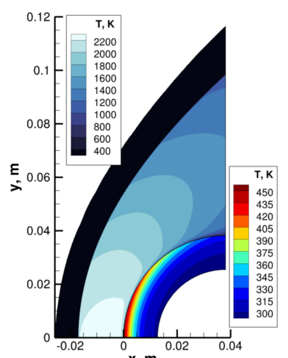
Research Spanning Hypersonic Physics & Applications



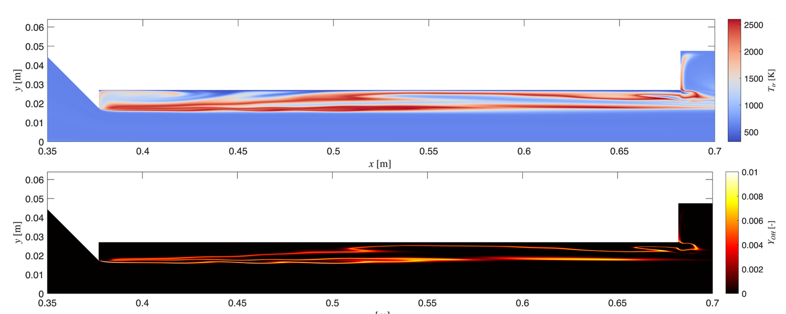
Plasma



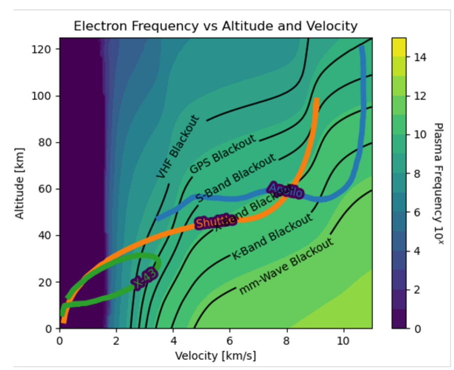
Rarefied Flows and Kinetic Approaches



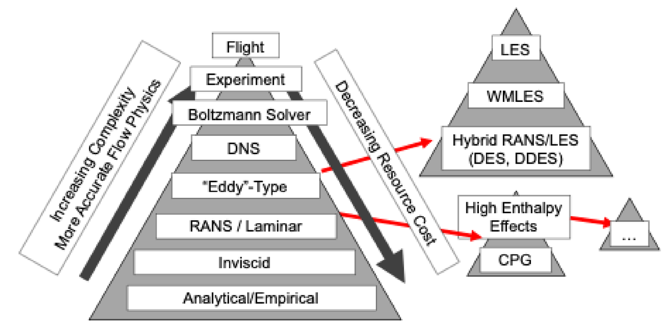
Reentry and Material Response



Propulsion



Aero-Optics and Blackout



Multi-Fidelity Approaches and Surrogate Modeling

Sponsors